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$y(x) := \sqrt{x}$

$Max32 := \left(2^{32} - 1\right)$

$clz(x) := \text{if } x = 0 \; 32 \text{ else } 32 - \text{trunc}\left(\log_2\left(\text{trunc}(x) + \frac{2^{32}-1}{2^{32}}\right) + \frac{2^{32}-1}{2^{32}}\right)$

$msb(x) := 32 - clz(x)$

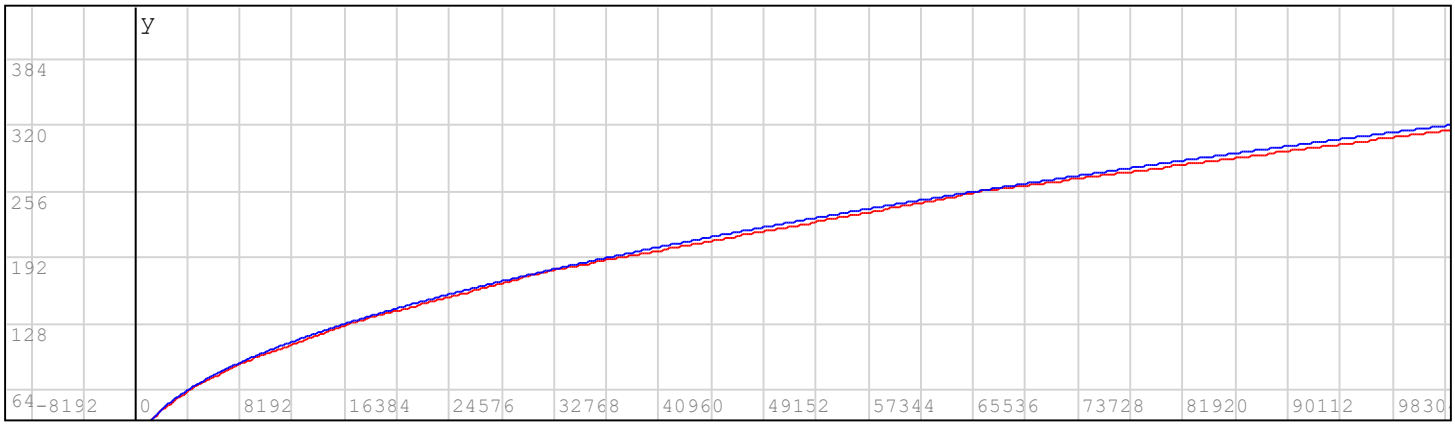
$$linScale(x) := \frac{\text{trunc}\left(\text{trunc}\left(\left(\sqrt{2}\right)^{msb(x)} - \left(\sqrt{2}\right)^{msb(x)-1}\right) \cdot Max32\right)}{2^{32}}$$

$$Xcorr(x) := \text{trunc}\left(\left(\frac{\text{trunc}(x) - 2^{msb(x)-1}}{2^{msb(x)-1}}\right) \cdot (Max32)\right)$$

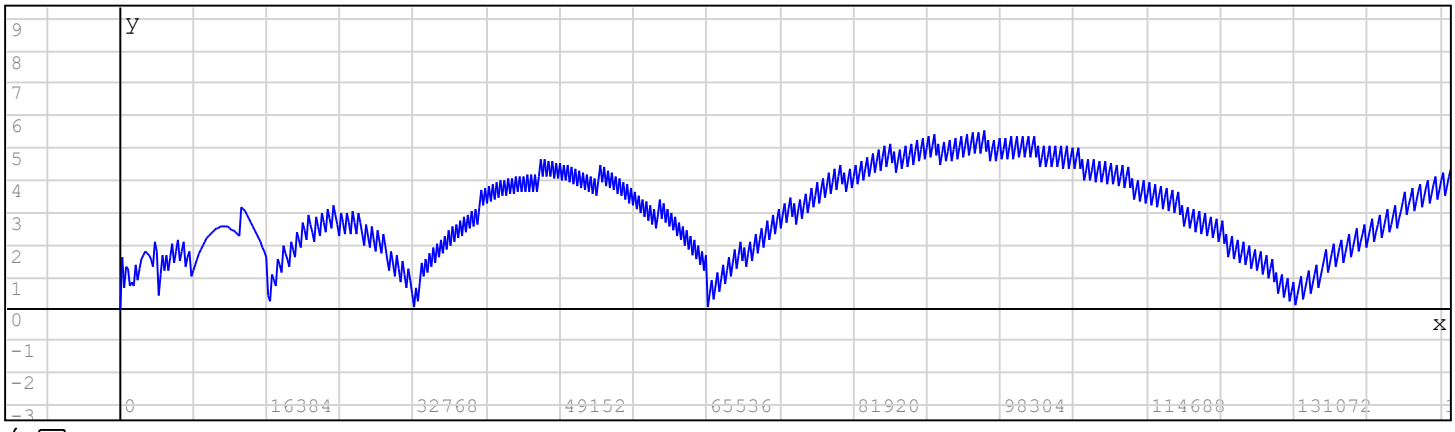
$$\text{trunc}\left(\left(2 - \sqrt{2}\right) \cdot 2^{32}\right) = 2515933592$$

$$\text{trunc}\left(\left(\sqrt{2} - 1\right) \cdot 2^{32}\right) \cdot 2 = 3558067406$$

$$linSqrt(x) := \text{trunc}\left(\frac{Xcorr(x)}{2^{32}} \cdot linScale(x)\right) + \text{trunc}\left(\sqrt{2^{msb(x)-1}}\right)$$




$\sqrt{x}$
$linSqrt(x)$



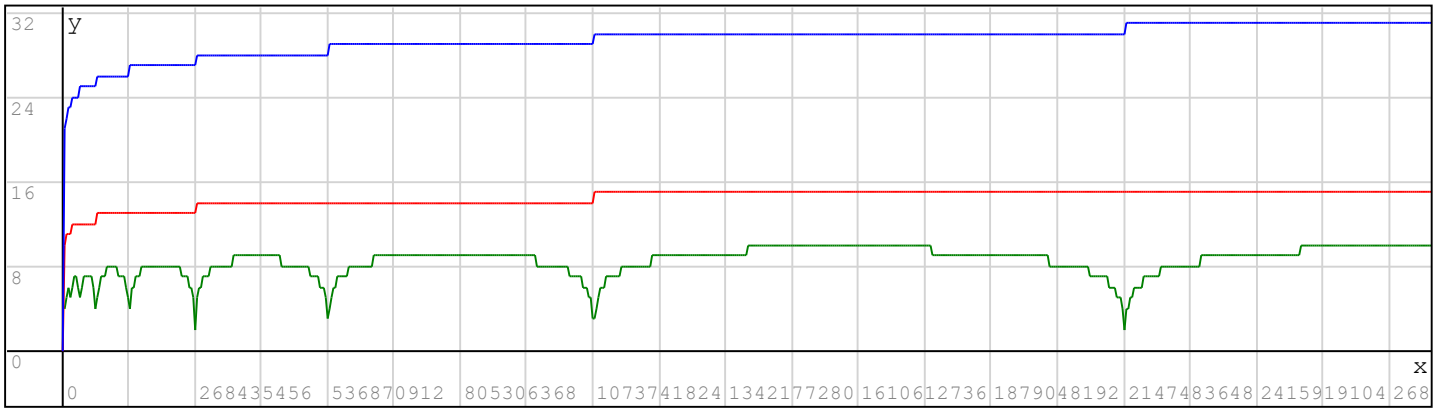

$\sqrt{x} - linSqrt(x)$

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$$argBitCount(x) := \text{trunc}\left(\log_2\left(x + \frac{2^{32}-1}{2^{32}}\right)\right) \qquad resultBitCount(x) := \text{trunc}\left(\log_2\left(\sqrt{\text{trunc}(x)} + \frac{Max32}{2^{32}}\right)\right)$$

$$linearBitError(x) := \text{trunc}\left(\log_2\left(\left(\sqrt{\text{trunc}(x)} - linSqrt(x)\right)\right) + \frac{Max32}{2^{32}}\right)$$




$$\begin{cases} argBitCount(x) \\ resultBitCount(x) \\ linearBitError(x) \end{cases}$$

$$linSqrtErr(x) := \sqrt{x} - linSqrt(x)$$

$$PI4 := \left(\frac{2 \cdot \pi}{2^3}\right)$$

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$$QXcorr(x) := \text{trunc}\left(\frac{Xcorr(x)\cdot Xcorr(x)}{2^{32}}\right)$$

$$TXcorr(x) := \text{trunc}\left(\frac{QXcorr(x)\cdot Xcorr(x)}{2^{32}}\right)$$

$$FXcorr(x) := \text{trunc}\left(\frac{TXcorr(x)\cdot Xcorr(x)}{2^{32}}\right)$$

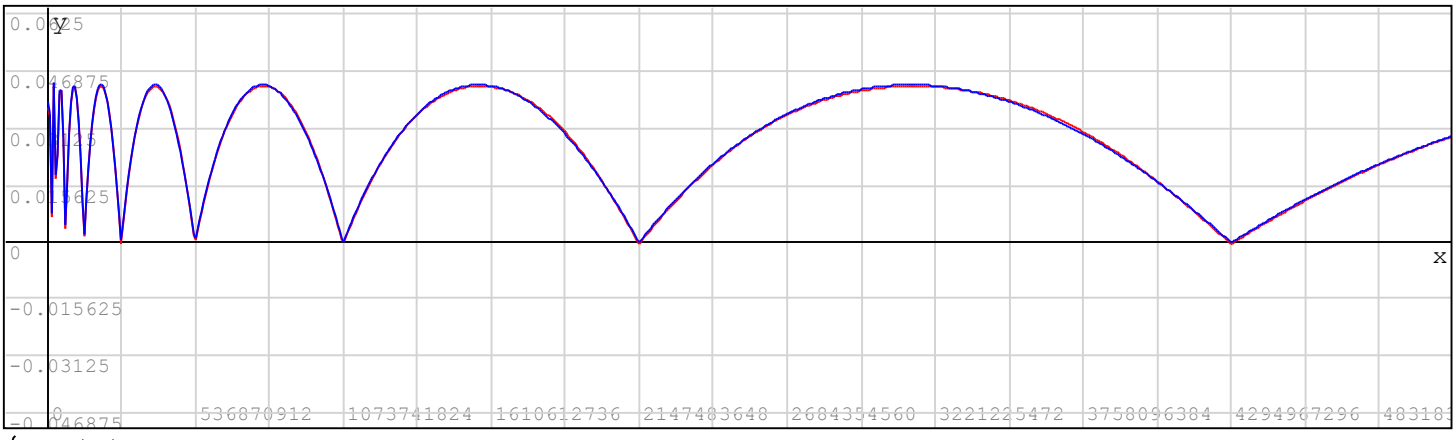
$$QK := \text{trunc}\left(Max32\cdot\frac{1.02035}{2}\right) \qquad TK := \text{trunc}(Max32\cdot 0.480) \qquad FK := \text{trunc}(Max32\cdot 0.52)$$

$$Xnew(x) := -\frac{Xcorr(x)}{2} + \text{trunc}\left(QXcorr(x)\cdot\frac{QK}{2^{32}}\right) + \text{trunc}\left(TXcorr(x)\cdot\frac{(TK)}{2^{32}\cdot 2^{2}}\right) - \text{trunc}\left(FXcorr(x)\cdot\frac{FK}{2^{32}\cdot 2^{2}}\right) + \text{tr}$$

$$Scale := \text{trunc}\left(\frac{Max32}{2^{5}}\cdot 16.48\right) \qquad Offs := 20$$

$$ec(x) := \text{trunc}\left(\frac{\text{trunc}\left(\sin\left(\frac{\text{trunc}(Xnew(x))}{2^{32}}\cdot\pi\cdot 2\right)\cdot Scale\right)}{2^{3}} - Offs\cdot 2^{16}\right)\cdot\frac{1}{2^{32}}$$

$$err(x) := \frac{linSqrtErr(x)}{linScale(x)}$$




$err(x)$
$ec(x)$

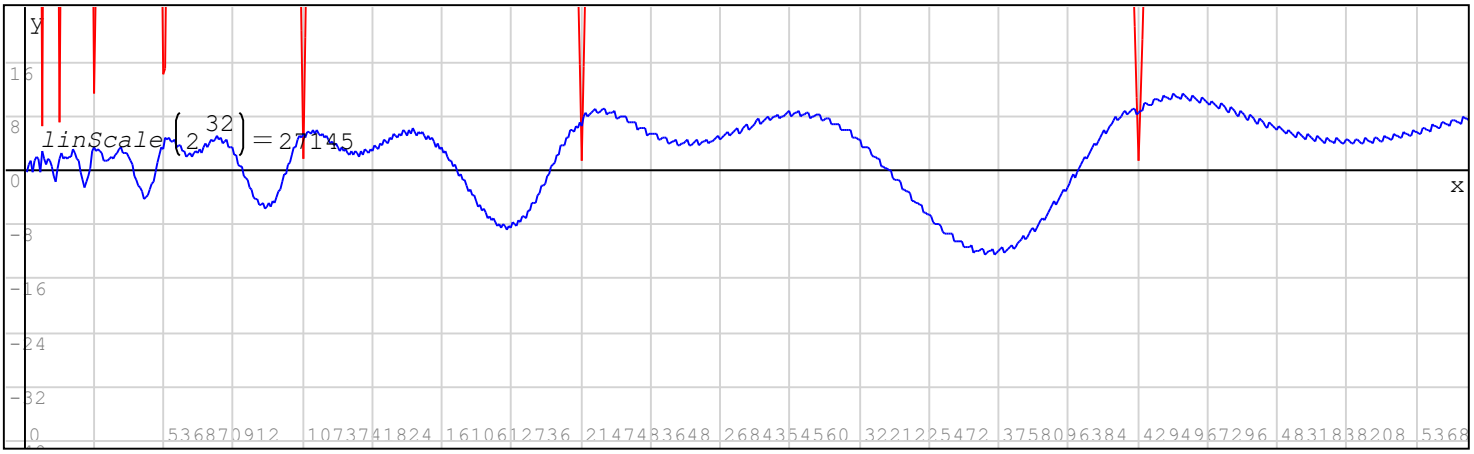
$QK = 2191184939$ $TK = 2061584301$ $Offs = 20$

$FK = 2233382993$ $Scale = 2211908156$

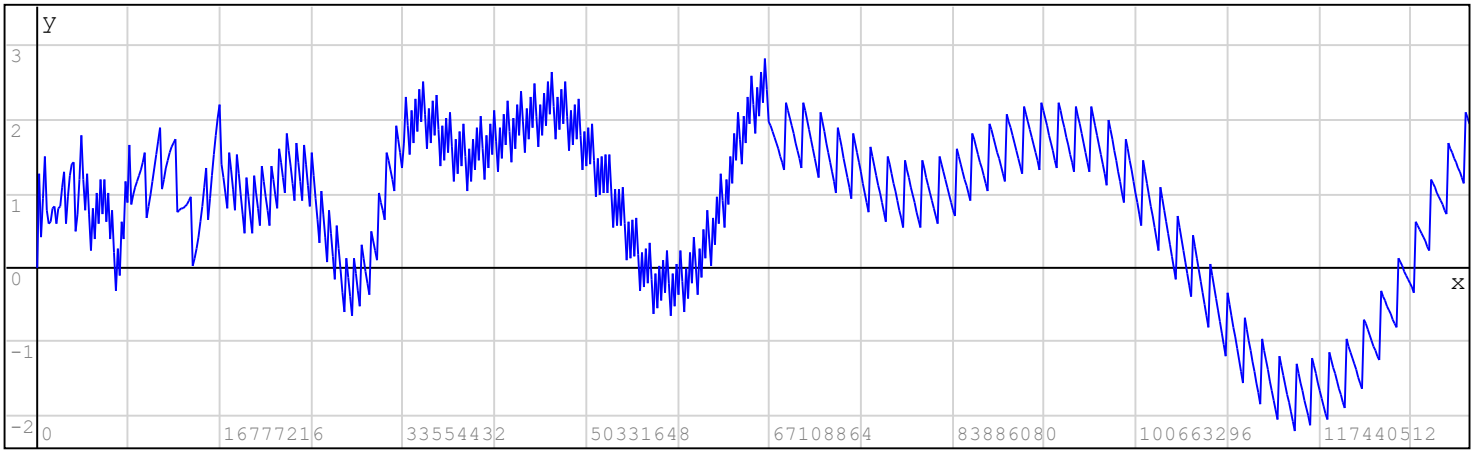
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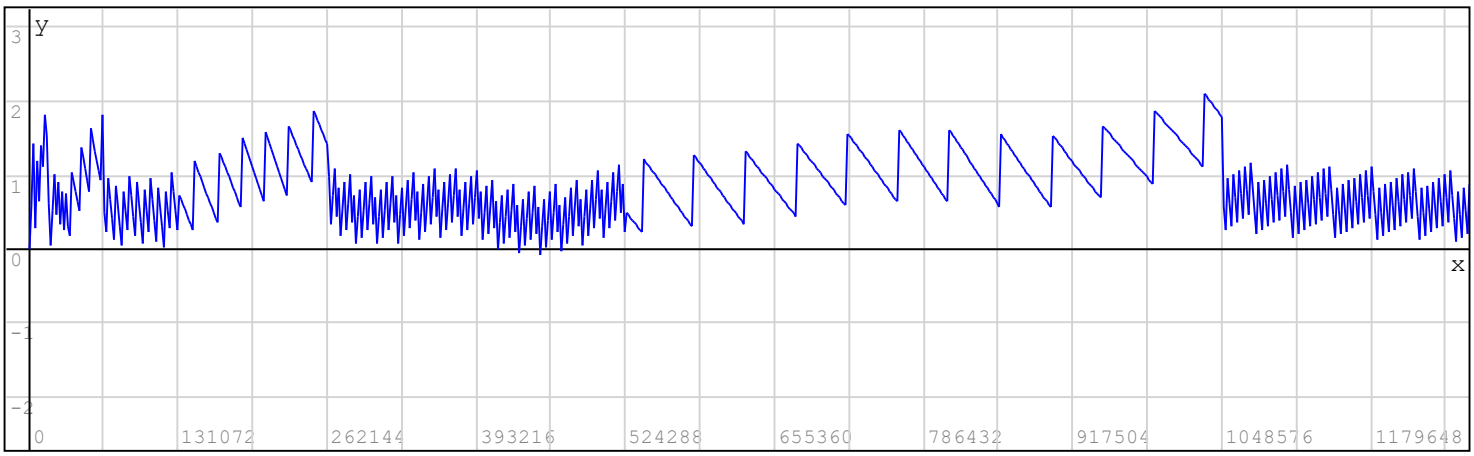
$cosSqrt(x) := (linSqrt(x) + (ec(x) \cdot linScale(x)))$




$\{\sqrt{x} - cosSqrt(x)$
$\{\sqrt{x} - linSqrt(x)$




$\{\sqrt{x} - cosSqrt(x)$




$\{\sqrt{x} - cosSqrt(x)$

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